

Backend Technology Stack

Chosen for correctness and developer speed at 100-customer scale, not for throughput.

Layer	Choice	Version / Package
CORE RUNTIME		
Language	Python	3.13 (3.14 if deps are clean)
Package manager	uv	latest
Web framework	FastAPI	latest
ASGI server	Uvicorn	with workers
Monorepo	uv workspaces	-
DATA LAYER		
Database	PostgreSQL	18
ORM	SQLAlchemy	2.0 async
DB driver	asyncpg	latest
Migrations	Alembic	latest
Cache / queue store	Redis	7+ (redis-py async)
Background jobs	ARQ	latest
Object storage	Cloudflare R2	via boto3
Vectors	pgvector	halfvec + HNSW
API AND CONTRACTS		
Validation	Pydantic	v2
Config	pydantic-settings	latest
API spec	FastAPI built-in	OpenAPI 3.1
Docs UI	Scalar	scalar-fastapi
Frontend client	orval	or openapi-typescript
AUTH		
Password hashing	argon2-cffi	latest
JWT	PyJWT	EdDSA, 15-min access
Refresh tokens	Custom	-
MFA	pyotp	TOTP + recovery codes
Permissions	Hand-rolled policy functions	-
Enterprise SSO	DEFER	-
COMMUNICATIONS AND DOCUMENTS		
Email	Resend or AWS SES	-
Email templates	Jinja2 + premailer	-
SMS	MSG91	-
WhatsApp	Meta Cloud API	direct or via BSP
PDF	WeasyPrint	latest
Search	Postgres FTS	tsvector + GIN, pg_trgm
QUALITY AND OBSERVABILITY		
Lint + format	Ruff	latest

Type checking	mypy	strict mode
Test runner	pytest	+ pytest-asyncio
Integration tests	testcontainers-python	real Postgres
API tests	httpx.AsyncClient	ASGI transport
Test fixtures	polyfactory	Pydantic-aware
E2E	Playwright	latest
Logging	structlog	JSON output
Tracing	OpenTelemetry	FastAPI + SQLAlchemy + asyncpg ins
Errors	Sentry	Python SDK
Telemetry backend	Grafana Cloud free tier	-
Internal admin	SQLAdmin	latest

Pin exact versions in uv.lock at kickoff. Node/Postgres/Python release lines move; the choices above are stab

Why this one
Mature production floor; strongest AI ecosystem
Replaces pip/poetry/pyenv; lockfile-first; makes Docker builds fast
Pydantic-native, auto OpenAPI 3.1, native async for LLM streaming
Standard; swap to Granian later if throughput demands
Native to uv; links local packages without extra tooling
RLS, JSONB, pgvector and full-text search in one engine. 19 is beta until ~Oct 2026
Typed ORM; transaction API exposes the raw connection needed for RLS
Fastest async Postgres driver
Autogenerate + hand-written RLS policy migrations
Cache, rate limits, job queue
Async-native, Redis-backed, built-in cron scheduler
S3-compatible, zero egress fees; move to S3 by changing one env var
Same database means tenant isolation comes free via RLS
Request/response models; also powers AI tool schemas
Typed env config validated at boot; crash on missing keys
Generated from Pydantic models automatically
Cleaner than Swagger UI
Generates a typed TS client from the OpenAPI spec
Use directly — passlib is effectively unmaintained
Well maintained; avoid python-jose
Rotating, stored hashed in Postgres, reuse detection revokes the family
Simple and sufficient
More readable and testable than a rules engine at this size
Add WorkOS or Keycloak when a customer pays for it
SES is cheapest at volume; Resend is simplest
CSS inlining for mail client support
India-native, handles DLT compliance, far cheaper than Twilio
AiSensy / Gupshup as BSP options
Pure Python, strong CSS Paged Media support; no extra Railway service
Outperforms bolt-ons at this scale and respects RLS for free
Replaces black, isort, flake8

Stability over speed
-
MANDATORY — RLS cannot be tested against a mock
-
-
-
Bind tenant_id, user_id, trace_id per request
-
-
Move to self-hosted SigNoz once on AWS
Covers ~80% of Django Admin for support work in ~100 lines of config

le but the numbers should be re-verified on day one.

Alternative considered
TypeScript / NestJS
Poetry, pip-tools
Django + DRF (better admin, weaker async)
Granian, Hypercorn
Poetry monorepo plugins
-
SQLModel (less mature)
psycopg3
-
-
Celery (if team knows it); procrastinate (drops Redis)
AWS S3, Backblaze B2
Qdrant (only past ~10M vectors)
-
-
-
ReDoc, Swagger UI
Hand-written client
bcrypt
joserfc
-
-
Casbin, Oso
-
Postmark
MJML
Twilio
-
Gotenberg container
Typesense, OpenSearch
-

pyright, ty
-
-
-
factory-boy
Cypress
-
-
-
Pydantic Logfire (single tool, data outside India)
Custom admin app